Global Facilities - Design & Construction Standard - Gas - Bulk Gas Supply and Distribution Systems (N2, O2, Ar, He, H2, CO2) Specifications

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1.0 Purpose

1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering and Construction of new bulk gas systems (N2, O2, Ar, He, H2, CO2) as part of new Greenfield FAB Construction projects

1.2 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

2.0 Scope

2.1 The document covers the new Basebuild System installations only and does NOT cover tool installation.

2.2 Site(s) impacted

This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects.

3.0 Definitions and Acronyms

Acronym	Definition / Description
Ar	Argon
ASU	Air Separation Unit
CDA	Clean Dry Air
CHW	Chilled Water
CO2	Carbon Dioxide
CQC	Continuous Quality Control
CW	Condenser Water
FM	Factory Mutual
FMCS/SCADA	Facility Monitoring and Control System/ Supervisory Control and Data Acquisition
H2	Hydrogen
He	Helium
ICA	Instrument Control Air
LAR	Liquid Argon
LH2	Liquid Hydrogen
LIN	Liquid Nitrogen
LOTO/COHE	Lockout Tagout / Control of Hazardous Energy
LOX	Liquid Oxygen
MAC	Main Air Compressor
N2	Nitrogen
O2	Oxygen
PLC	Programmable Logic Controller
PPB	Part per billion
PPM	Part per million
PPUPPU – Pre	Pre-Purification Unit Purification Unit
TUM	Tool Utility Matrix
UPS	Uninterruptible Power Supply
VSD/VFD	Variable Speed Drive / Variable Frequency Drive

4.0 Roles and Responsibilities

Role	Responsibilities
Global Facilities Engineering & Regional Team	• Develop and maintain document (Base Build aspect) • Ensure Best Known Methods (BKM), lessons learned, are documented in document. • Perform periodic document review and update document • Review and update scope delineation of new system when introduce and recommend new Facilities system for both new Greenfield FAB Construction projects, and for existing Site upgrade projects. • Review and provide technical feedback on document • Support Site teams by providing knowledge and technical expertise. • Support the Site teams by providing knowledge and technical expertise. • Review and update scope delineation of new system when introduce and recommend new Facilities system for existing Site upgrade projects.
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to Global Facilities Engineering & Regional Team Representatives, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard.
Site Facilities Team	• Use System Standard as technical reference • Identify and document deviations from the system standard/specification for review and approval by Global Facilities Engineering & Regional Team Representatives.

5.0 Overview of Systems

5.1 Type of supply system

Bulk gases such as N2, O2, Ar, He, H2, and CO2 are supplied to the fab to support manufacturing.

The gases can be delivered from a cryogenic source or high-pressure gas source. Bulk deliveries supplied as liquid because it requires much less storage capacity than gas form. Typically for liquid/cryogenic gas supply, the scheme normally consists of a liquid/cryogenic source, followed by a vaporizer which vaporize the liquid gas to pressurized gas to be supplied downstream. This followed by a downstream panel consisting of valve manifold and filter to regulate down the pressure to required setpoints before supplying to the fab.

If the volume of the gas required is relatively low, the supply mode will be from a high-pressure gas source. Typically, the high-pressure source is connected to a downstream panel consisting of valve manifold and filter to regulate down the pressure to required setpoints before supplying to the fab.

Depending on the pressure, volume and flow rate requirement, the most appropriate and cost-effective gas supply mode can be identified. Below is the available packaging available for bulk gas supply ranging from lowest to highest supply volume:

a. High pressure gas : Cylinder
b. High pressure gas : Pallet/Bundle
c. High pressure gas : Trailer (20ft/40ft)
d. Liquified gas : Dewar / Liquid Gas Cylinder (LGC)
e. Liquefied gas : Vacuum insulated bulk storage tank

Preliminary design and guidelines shall be submitted to Factory Mutual (FM), and recommendations shall be evaluated during the design phase. This is applicable for O2 services and H2 services.

5.2 Supply system package

5.2.1 Liquid tanks

A dedicated space with civil requirement is required to erect the storage tank on-site, consideration to be taken to allocate the access pad for the tanker filling as well is needed. The liquid gas is delivered/transferred by cryogenic liquid tankers into the vacuum insulated bulk storage tank on-site. Typically, the bulk storage tank on-site is owned and maintained by the gas suppliers. It can be fitted with telemetry systems to provide real time remove monitoring of the stock/content levels by the gas suppliers to ensure logistic and product is always available.

5.2.2 Trailers / ISO tanks

A dedicated space with civil requirement is required to park and allocate trailer/isotanker on-site, consideration to be taken for the route and logistic of the supply package. It can be fitted with telemetry systems to provide real time remove monitoring of the stock/content levels by the gas suppliers to ensure logistic and product is always available.

5.2.3 Pallet / ton containers

Pallet/ton containers provide a larger volume of gas supply than single cylinder. This will reduce the rate of cylinder change-out required in term of operation. Below consideration of upgrading/upsizing the supply package;

i. Change-out shall be maintained to a minimum of 3 days/package size to allow for flexibility and troubleshooting during change-out
ii. Logistic and supply from gas supplier. Team to consider the delivery distance from supplier to Micron site and weather conditions in specific sites. A higher storage volume is required to avoid any complications in delivery.
iii. On-site storage requirement

5.3 Supply system sizing

At a minimum, the system's capacity to supply the bulk gases shall be sized for 30% more than the projected demand/forecast at full ramp. Selection of the size, quantity, and components shall also consider the following:

a. Manufacturer's standard equipment sizing and capacity offering the highest energy efficiency and/or lowest capital cost;
b. Available area for supply system

On site storage volume shall be sized to accommodate a minimum of 3 days forecasted maximum flow.

Description	Parameters
On-site Storage (days) @ forecast maximum flow	A
# haul in/frequency supplier commitment/ capability	B
# haul in/frequency required @ forecast maximum flow	C

On-site storage: Defined as the equivalent gas volume being stored onsite in the storage tanks (for liquid), or supply package e.g. trailers, ISO tanks and etc. This volume will be ready to supply for manufacturing without considering haul-in or package change. On-site storage (days) @ forecast maximum flow (A) : A>3 days

Examples:

i. With the maximum flow of 100Nm3/h of O2, the minimum required size of liquid tank on-site should have an equivalent of (100Nm3/h x 24h x 3 days) amount of gas. This translates and enables the site to have the capacity to supply at maximum flow without the need from external liquid haul in.

ii. With the maximum flow of 100Nm3/h of H2, the minimum required size of a supply package/trailer should have an equivalent of (100Nm3/h x 24h x 3 days). This translates and enables the site to have the capacity to supply at maximum flow without the need to switch or change-out the supply package/trailer.

Haul in : B>C (additional mitigation and provision need to be considered if this can't be met. Logistics and gas supplier BCP for Micron to be reviewed and accessed)

5.4 System configuration

5.4.1 Number of supply/standby/backup system

System shall have a primary supply system with a secondary system (similar capacity) for redundancy and backup. E.g.;

i. For liquid supply by vacuum insulated bulk storage tank, design shall cater for a minimum of 2 number of storage tank to provide redundancy and backup
ii. For supply scheme with panel with A&B, system C (as minimum) shall be designed to cater for redundancy and backup. The A&B system will be considered as primary supply and The C system will act as secondary supply.

5.4.2 Capacity sizing

At a minimum, the system's capacity to supply the bulk gases shall be sized for 30% more than the projected demand/forecast at full ramp.

All component shall be documented with nominal and maximum flowrate to identify the bottleneck and limiting component for future evaluation and expansion.

5.4.3 Redundancy requirement

All components shall be design with redundancy in mind and single point of failure (SPOF) analysis shall be done.

If redundancy of components can't be done due to cost, space, shelf life factors; sufficient instrument/components spare parts shall be available for change-over with no or minimum downtime. Critical spare part shall be available and training on emergency response and change-out shall be provided.

5.5 Bulk Gas BCP Guideline and Criteria for Greenfield Projects

At a minimum, all greenfield construction should follow the guidelines shown below for new gas plants. In geographies, where raw material sourcing is harder, supplier should work with Micron to define minimum backup volume so as to ensure no impact to Micron process continuity.

Criteria	Current Guidelines for Greenfield Project (N_2) (based on design standard [DS], business continuity plan [BCP], operational practice [OP])
Gas plants capacity	• [DS] Minimum 30% more of projected demand at full ramp. • [OP] Minimum 2 units of Gas plant for redundancy to cater the total capacity.
On-site storage inventory	• [BCP] Minimum 3 days backup equivalent volume, based on 2 largest gas plants capacity. • [BCP] Minimum 3 days backup equivalent volume, based on forecast maximum flow.
Supplier Inventory	• [DS] Haul-in/day supplier commitment > haul-in/day required with largest ASU failure. • [DS] Continuous haul-in (>15days) committed by supplier.
System Reliability	• [DS] N+1 filter & compressor. • [DS] Different operation modes (50%, 80% & 100%) of gas plants. • [DS] Critical equipment spare part management. • [DS] Segregation by power source on N+1 major equipment.
Contamination	• [DS] Design spec alignment & CQC analyzer

5.6 Main component description and requirement

5.6.1 Vaporizer

Types of vaporizers:

i. Ambient vaporizer – Natural Draft / Forced Draft
ii. Water bath vaporizers – Indirect heat by combustion gases, electric heater or steam
iii. Water circulating vaporizers
iv. Electric vaporizers
v. Combination of above

For vaporizer type/design which require switchover between online and off-line units when operational units are saturated; the vaporizer shall have redundant capacity and sized two (2) times 100% capacity.

For vaporizer type/design which doesn't require switchover between units and capable of running 24/7; the vaporizer shall have N+1 configuration.

Vaporizer shall be designed for a continuous maximum sustainable flow rate. If de-frosting is required at a given frequency, the vaporizers' capacity shall be adequately increased to ensure an uninterrupted flow rate service at maximum capacity.

Vaporizer capacity should be sized for the worst 10 year weather condition at installation site. When the gas leaves the vaporizers, required specification must be able to be met while running 24/7 continuously.

Temperature will be modeled based on outlet T from vaporizers, pipeline routing and external temperature. The minimum temperature entering the site building will be greater or equal to 5°C, to avoid condensation from main pipeline inside the Fab building. Insulation and appropriate heating shall be considered.

Supplier shall recommend a model or method of vaporizer operation based upon vaporizer capacity and duty cycle and include rotation to allow for sufficient de-icing for extended use. Temperature monitoring/interlocks shall be included post vaporizer where necessary and auto switchover on timer function where multiples vaporizers is installed in order to provide rest and recovery period for redundant vaporizer.
Each vaporizer bank must have capability of being isolated.

5.6.2 Gas Panel

Panel shall be designed with auto switchover function with signal/status tied and monitored in Micron SCADA system.

Gas panel located at the gas yard shall be rated for outdoor usage;

i. A minimum IP rating of IP55 for enclosure housing electrical and mechanical components
ii. SS hard tubing shall be used for pneumatic air supply where possible and UV rated pneumatic air tubing
iii. All pneumatic non-metal tubing shall be (PFA (Perfluoroalkoxy)) iv. Shaded and protection from direct sunlight and weather elements

Hydrogen shall be treated as HPM for its flammability characteristics. Hydrogen shall be distributed whereby with mechanical fitting indoors must be in exhausted and monitored enclosures. Outdoor mechanical fitting does not need to be enclosed, exhausted, and monitored. Please refer details on the distribution node on Specialty Gas System Standard.

5.6.3 Pressure monitoring

Pressure monitoring shall be provided to monitor inlet (source), delivery pressure to the fab and end of the line. All analog signal shall be sent and monitored at the Gas Monitoring System (GMS)/Micron SCADA system

Lower limit and upper limit shall be set to trigger warning to the operator via GMS/SCADA.

POC – Delivery pressure Point of Connection delivery pressure. It is the minimum inlet pressure at the Tool boundary in Psig; it is identified at the system start up by Operation and tool owner, and normally documented in site data base. For bulk system delivered by laterals; a minimum value shall be picked and discussed with the tool owners for a consensus. Further downstream regulator to be installed at the tool to further reduce the pressure to required tool’s setpoint.

LPOC – Delivery pressure Lateral Point of Connection delivery pressure. It is the most appropriate delivery pressure in Psig of the LPOC valve, of the system distribution; it can be measured at the LPOC valve, at the VMB stick, or any other valve having the same function, located in the distribution, based on the system configuration.

Fluctuation (±) is intended as the acceptable operating range, as a result of distribution sizing, tool requirements, and components performances.

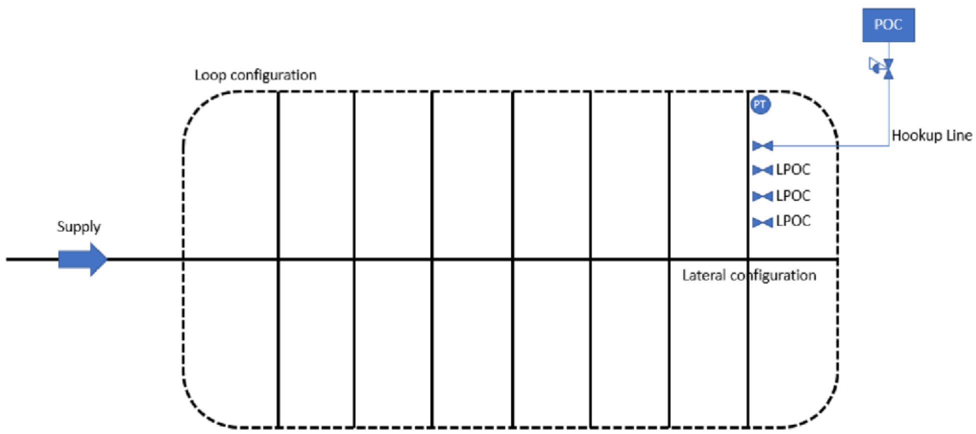
LPOC L alarm - Minimum LPOC GMS/SCADA low pressure alarm threshold. It is the most appropriate Low delivery alarm pressure threshold of the LPOC, in the event of system failure. Principle is based on a threshold value lower than typical fluctuation to avoid/minimize false alarms. This is enough yet to supply the tool and allow Operation to troubleshoot alarm preventing interruptions. This parameter can be monitored with a pressure transmitter end of line at the LPOC location.

LPOC H alarm - Maximum LPOC GMS/SCADA high pressure alarm threshold. It is the most appropriate High delivery alarm pressure threshold of the LPOC, in the event of system failure. Principle is based on a threshold value higher than typical fluctuation to avoid/minimize false alarms. This is enough yet to supply the tool and allow Operation to troubleshoot alarm preventing interruptions. With this parameter, any intrusion of higher pressure gas or abnormalities at the gas source can be alerted and detected. This parameter can be monitored with a pressure transmitter end of line at the LPOC location.

Schematic and guide for the parameters are as below;

Parameters	Value	Remarks
POC	-	Information from tool owners and required tools
Fluctuation	± 5 Psig	Value is selected based on instrument sensitivity/repeatability (+/-1.0% Reading accuracy), design tolerance shall be +/-2 Psig, tolerance range from tools and response time from operation
LPOC	LPOC = POC + 15 Psig (minimum)	Additional of 15 Psig at the LPOC before supplying to tool; will enable pressure drop downstream at piping and components. With a gap of 15 Psig, the pressure at the tool can be regulated effectively with downstream regulator
LPOC L alarm	LPOC – Fluctuation	The alarm is to capture any abnormal readings during the operation and possible instrument drift over time

LPOC H alarm	LPOC + Fluctuation	The alarm is to capture any abnormal readings during the operation and possible instrument drift over time



Example
Gas/medium : Oxygen, O2
POC : 80 Psig (input from tool owner)
Fluctuation : ± 5 Psig
LPOC : 80 + 15 Psig = 95 Psig
LPOC L alarm : 95 – 5 Psig = 90 Psig
LPOC H alarm : 95 + 5 Psig = 100 Psig

The distribution network shall be designed to a limiting total pressure drop of 2 psi across the network. Pipe sizing, component sizing, and configuration (lateral design, or loop design) shall be designed to achieve the required delta P.

The LPOC and all valves shall be lock out tagged out to appropriate position after testing and commissioning and during operation. This will prevent mis-operation or accidental operation of the valves during operation.

5.6.4 Regulator

Reference for APTech regulator and valve selection based on gas type, flow required and etc can be found in the Pressure Regulator and Valve Selection Guide. Regulator Seat Abrasion by APTech can be found Pressure Regulator Seat Abrasion Product Note

i. Fixed regulator

Regulator shall be selected based on several criteria;

a. Gas type – corrosive gas might need special material or diaphragm, smaller molecule gas e.g. H2, He might prone to regulator passing
b. Flowrate – high flow model of regulator is needed for high consumption, Cv of the regulator will be taken into consideration. For most of the case, regulator will be the bottleneck and restriction for the maximum flow for the system
c. Inlet and required delivery pressure tolerance – Inlet pressure variation will be huge for high pressure gas as the switchover point for the source is recommended at 10% of full supply pressure. The maximum allowable working pressure (MAWB) for the regulator shall be rated to accommodate this. Downstream delivery pressure tolerance need to be identified e.g. downstream high pressure will be caused by supply pressure effect (SPE) or regulator creep, whereby downstream low pressure will be

caused by undersized of regulator with low required Cv and with high flow condition.

d. Supply Pressure Effect (SPE) or drooping

e. This is a general characteristic for regulator. Effect can be minimized by dual stage regulator with flow capacity tradeoff.

ii. Double stage regulator

In general, double stage regulator is used to eliminate the SPE of single stage regulator; whereby a precise and stable downstream pressure is required.

In addition, double stage regulator will reduce the thermal expansion or Joule-Thomson effect. In B2H6 case, which affecting the decomposition and also the cleanliness of the gas.

Reference and information of APTech two stage regulators can be found in the 2-Stage Regulators Product Note. Supply Pressure Effect (SPE) on regulator and information from APTech can be found Supply Pressure Effect Concerns for High-Flow Regulators Used in High-Pressure, Compressed Gas Applications Product Note.

iii. Pneumatic actuated regulator

Pneumatic actuated regulator shall be used and considered if any conventional regulator can't meet the requirement. This is especially on special application with high flow where conventional regulator has a fixed lower Cv. Inconsistent flow and intermittent peak flow will cause maintaining stable pressure downstream difficult with conventional regulator. A Proportional Integrated Derivative (PID) controller coupled with pneumatic actuated regulator can handle a with range of flowrate within a tight pressure setpoint. The system shall be commissioned and tested out with the same range of flowrate before allowed for operation. This is to ensure the setting and response for PID is within tolerance limit and suited for the operation.

5.6.5 Block heater

Heater is required to increase the inlet temperature of the gas upstream regulator with Joule-Thomson effect and to prevent condensation/icing at the outlet piping surface of the regulator.

Heating element shall have more than one temperature sensor/cut-off for redundancy in preventing overheating. Heating element shall have thermostat/auto cut-off in the event of overheating. Heating and insulation shall not be flammable in nature and compatible with the application.

The power source for the heater shall be Normal or Emergency (single). Refer guide by Global Facilities – Electrical – Electrical Specification Design Standard

5.6.6 Line heating

Line heating is required for low vapor pressure gas; to prevent re-condensation along the distribution line due to temperature change, and high flow.

Line heating is required for special application where temperature of the gas need to be controlled within a certain range. Critical phases of the gases to be maintained as well in correlation in pressure and temperature equilibrium.

Heating element shall have more that 1 temperature sensor/cutoff for redundancy in preventing overheating. Heating element shall have thermostat/auto cut-off in the event of overheating. Heating and insulation shall not be flammable in nature and compatible with the application.

5.6.7 Safety components

i. Relief valve

Relief valve is referred as the mechanical protection along the gas distribution in protecting component and personnel against high pressure in the event of emergency. Primary protection is from interlock or warning/shutdown from pressure monitoring along the gas distribution.

Relief valve and burst disc shall be installed accordance to the local and international code requirement. Dual relief valve shall be applied to enable continuous protection of the system during the maintenance or calibration of the component.

For cryogenic application, relief valve shall be considered in each section whereby pressure build-up due to vaporization of liquid/trapped liquid. The pressure relief system shall guarantee the pressure will never exceed the pipe and component's design pressure. Consideration to be taken especially for pipe end where liquid might be trapped and a relieve mechanism to be in place.

Below criteria shall be considered in the selection for the relief valve/burst disc;

a. Material compatibility (surface finish, operating temperature, material)
b. Rating (refer below)
c. Capacity / vent volume – the relief valve/burst disc shall be able to relieve the excessive volume and reduce the pressure of the system to safe level in the event of emergency. The size of the valve/burst disc, downstream component and piping shall be adequate to channel the high-pressure gas to safe area within reasonable time

Below criteria shall be considered in the rating (set pressure) for the relief valve/burst disc;

a. Shall not exceed the maximum allowable working pressure (MAWP) of the weakest component in the distribution. The valve must open at or below the MAWP of the component in the distribution to prevent failure or damage to the component. The MAWP of the component should be at least 10% greater than the highest expected operating pressure under normal circumstances. This setting shall be staggered and aligned with other protection device e.g. pressure monitoring interlocks, or burst disc and relief valve
b. Temperature effect should come into consideration for cryogenic application or supercritical gases condition.
c. All vent of the relief valve shall be to safe location. Back pressure; the vent outlet pressure should be considered as wrong sizing will cause the relief valve to pop open repeatedly and, hence damaging the valve.

ii. Burst disc

Burst disc will act as 2nd mechanical protection along the gas distribution in protecting component and personnel against high pressure in the event of emergency after relief valve. This will act as final defense line against high pressure hazard along the gas distribution line.

Burst disc shall come with two sets to enable periodic calibration.

A three-way valve upstream the burst disc to be installed to enable redundancy during trouble or calibration.

Relief valve and burst disc shall be installed accordance to the local and international code requirement.

For high purity application, the distribution line will be expected to be contaminated in the event of burst disc rupture as impurities from the environment will be able to seep into the distribution via the ruptured burst disc.

Below criteria shall be considered in the selection for the relief valve/burst disc;

a. Material compatibility (surface finish, operating temperature, material)
b. Rating (refer below)
c. Capacity / vent volume – the relief valve/burst disc shall be able to relieve the excessive volume and reduce the pressure of the system to safe level in the event of emergency. The size of the valve/burst disc, downstream component and piping shall be adequate to channel the high-pressure gas to safe area within reasonable time

Below criteria shall be considered in the rating (set pressure) for the relief valve/burst disc;

a. Shall not exceed the maximum allowable working pressure (MAWP) of the weakest component in the distribution. The valve must open at or below the MAWP of the component in the distribution to prevent failure or damage to the component. The MAWP of the component should be at least 10% greater than the highest expected operating pressure under normal circumstances. This setting shall be staggered and aligned with other protection device e.g. pressure monitoring interlocks, or burst disc and relief valve
b. Temperature effect should come into consideration for cryogenic application or supercritical gases condition.
c. All vent of the relief valve shall be to safe location. Back pressure; the vent outlet pressure should be considered as wrong sizing will cause the relief valve to pop open repeatedly and, hence damaging the valve.

iii. Automated shutoff valve

The requirement for interlock/automated shutoff valve upstream gas supply can be found in the following EHS document, Global EHS - Toxic Gas Monitoring and Double Containment Standard.

Required automated shutoff valve shall be installed accordance to the local and international code requirement. Valve configuration shall be designed “fail-safe”

The automated shutoff valve should be fail-safe or normally closed. The automated shutoff valve shall be pneumatic actuated. Supply of shutdown signal and supply air shall be redundant in order to prevent unnecessary interruption.

iv. Excess flow switch (EFS)

Excess flow switch shall be installed to cut-out the supply source in the event of high flow either caused by;

a. Downstream line break causing massive leak
b. Excessive flow due to mis-operation or massive leak
c. Abnormal high flow from manufacturing

Excess flow switch shall be sized accordingly to;

a. Excessive flow potentially causing hazard as above to be captured
b. Undersized excess flow switch can cause unnecessary shutdown and nuisance alarm

Refer to the Flow Trip Point Calculator from APTech for sizing of excess flow switch

EFS is required for Hydrogen services. Excess flow switches shall set with an adequate delay time with alarm configuration to prevent interruption in the event of a false alarm.

5.6.8 Flowmeter

Flowmeter along the gas distribution primarily installed for monitoring and charge/cost with gas supplier. Consideration as below in flowmeter selection:

a. Material compatibility, surface finishes and standard specification e.g accuracy, range, and repeatability
b. Non-intrusion type is preferable for monitoring purpose
c. Mass flow meter is preferable for charge/cost with gas supplier
d. Total cost of ownership to be considered including; sensor replacement, maintenance and consumables

Flowmeter installed shall be linked to the GMS/SCADA for continuous monitoring together with pressure and temperature.

5.6.9 Purifier and filter requirement

Purifiers may be required to meet Micron's bulk gas purity specifications as delivered to the facility and fab. Selection of purifier shall meet the required Micron specifications at POC.

Refer to the Global Facilities – General – Design Parameters Best Known Methods (BKM) for the latest required specification.

Gas supplier shall meet the minimum required specification at the outlet of purifier and at the same time provide the maximum impurities level at the inlet (with achievable outlet specifications). The same information to be documented for reference and consideration in the event of specification change at outlet or different source/incoming specification change.

Purifiers shall be sized to accommodate minimum 30% more than the projected demand/forecast of the bulk gas at full buildout, and N+1 purifiers shall be specified where N is the number of purifiers needed to achieve the projected demand/forecast including the minimum 30% overcapacity. Purifiers may be installed in phases to accommodate a gradual ramp up in phases. The purifier manifold shall be designed to allow installation of additional purifiers without interruption to the fab, and N+1 purifiers shall be maintained at all times as purifiers are added in phases.

For non-regenerative purifiers, the life of the purifier bed must be calculated and monitored each quarter (if not monthly). New beds must be ordered based on the earlier of the two conditions:

a. Lifetime of the bed is less than 30%

b. Time to end of life is less than 2 times lead time for a replacement purifier

Utilities e.g cooling tower, exhaust, and purge and regenerative gases to enable purifier to work normally shall be maintained and to have redundancy to prevent interruption to the purifier's operation.

i. Material

Selection of purifier and filter largely depends on the incoming contaminants and the required outlet purity

Incoming contaminants typically originated from the gas supplier; additionally, for corrosive gases, contamination can be picked up in the gas stream as a result of corrosion of the delivery system and tool

In general, contaminant classes include moisture, acids, bases, refractory compounds, and organic molecules (Table below), but contaminants can also develop from the process and reaction. For example, if oxygen (O_2) is inadvertently introduced during the silicon (Si) ingot growth process, the resulting silicon dioxide (SiO_2) deposition will cause defects before wafer processing even begins.

CONTAMINANT TYPE	STANDARD SURROGATE	TYPICAL IMPURITY LEVELS (OUTLET PURITY)
Acids NO_x, SO_x, Organic acids	SO_2	0.1 – 1 ppmV (<1 pptV)
Atmospherics CO_2, CO, O_2, N_2	CO_2, CO, O_2, N_2	0.1 – 1.0 ppmV (<100 pptV)
Bases Amines, organoamines, silazanes	Ammonia (NH_3)	1 – 100 ppbV (<1 ppbV)
Metals Fe, Mn, Mo, Ni, Cr, others	E34 list, Fe, Cr	1 – 100 ppbV (<1 ppbV)
Moisture	H_2O	0.1 – 25 ppmV (< 1 ppbV)
Organics Condensable (45 – 100 amu; butane, IPA, toluene) Non-Condensable (>100 amu; decane)	Toluene Decane	0.1 – 2 ppmV (<1 pptv)
Refractory compounds Halogen-containing HCs, silicon compounds	HMDSO	<100 ppbV (<1 pptV)

Purifiers are able to remove molecular contaminants, such as H_2O, O_2, CO_2, hydrocarbons, and others.

Gas purification also employs two primary mechanisms. Physical adsorption (or physisorption) involves the physical interaction between the impurity being removed and the adsorbent. For example, physisorption is used to remove H_2O and hydrocarbons. Chemical adsorption (or chemisorption) involves forming a chemical bond between surface atoms of the adsorbent and the impurity. This is how reactive oxides like CO and O_2 are removed. Below are list of absorbent classes and types available;

*List of adsorbent classes and types

ADSORBENT: POROUS MATERIAL CLASSES	ADSORBENT TYPES
Crystalline materials Exhibit reproducible x-ray diffraction patterns Class includes zeolites, metal-organic frameworks (MOFs), ZIFs, ALPOs, SAPOs, some forms of carbon (i.e., CNTs, graphene) Vast array of structure types provides range of separation selectivities	**Activated charcoal (carbon)** Physical adsorption Removal of organic and strong acid AMC
Regular materials Do not exhibit reproducible diffraction (amorphous) Class includes activated carbons, amorphous silicas, and aluminas Can be modified to enable/enhance separations	**Coated carbons** Combination physical/chemical adsorption Removal of weak acids, special chemistry also some organic AMC
Catalytic and gettering materials Typically, a formulation that contains transition metals and alumina or silica binder Primarily used for removal of oxygen, oxidation of VOCs	**Ion exchange media** Chemical adsorption Base and acid AMC
	Catalysts Destructive adsorption/oxidation Special materials and formulations

A filter is able to remove suspended particles from gas streams.

There are two dominant mechanisms in gas filtration. The first is inertial impaction, which is used when the particle is large and dense enough to pass out of the fluid streamline due to its inertia. The second mechanism is diffusion interception, which captures particles that are small enough to pass out of the fluid streamline due to Brownian motion. Factoring both, diffusion and interception can capture even the smallest particles, (about 0.001 µm or less).

Particle filtration should be 99.99999% removal of 0.003 micron particles for ultra high purity process gases (SEMI E49.7-0304 for N2 used for testing, drying and purging)

Particle filtration should be 0.05 micron retention particle sizes as minimum for high purity non-process gases. ii. Sizing

Purifiers shall be designed for a continuous maximum sustainable flow rate. If regeneration is required at a given frequency, the purifiers' capacity shall be adequately increased to ensure an uninterrupted flow rate service at maximum capacity. N+1 redundancy shall be provided as minimum. Pressure drop across the purifier shall not be more than 2.0 bar.

Filter skids shall be designed with redundancy, independently operated filter units having a 100% capacity as a minimum. Each filter unit shall be equipped with pre/post isolation valves and pressure transducers able to be interfaced with the Micron SCADA system and pressure gauge on common filter outlets.

5.7 Alarm and monitoring

Panels status should be monitored at the Micron SCADA.

Filter should have below for monitoring at the Micron SCADA;

a) Incoming pressure
b) Outgoing pressure
c) Differential inlet and outlet pressure of the filter

Purifier should have below for monitoring at the Micron SCADA;

a) Incoming pressure
b) Outgoing pressure
c) Mass flowmeter reading
d) Status of the purifier e.g. supplying, regenerating, standby, alarm and etc
e) Analyzer reading (if applicable)
f) Heater temperature (if applicable)
g) Interlocks (over temperature or temperature abnormalities shutdown, over pressurization alarm and interlocks)
h) Battery backup for PLC power

5.8 Purifier Room

a. Room area monitoring requirements (O_2, H_2)

Purifier/Filter room or gas room supplying inert gas shall be installed and monitored with Oxygen level detector. Alarms and escalation in the event of emergency or leak to be identified and informed to the operation team.

For H_2 services, detectors at possible leak points shall be installed to stop the gas source in the event of leak.

5.9 Power Requirement

The electrical system components shall have adequate withstand ratings based on the available fault current at the point of common coupling. The electrical system shall be selectively coordinated, including any over current devices upstream from the point of common coupling. Voltage and current harmonic distortion shall not exceed limits established in the latest edition of IEEE 519. Newer values will be dependent on the current consumption of the load at the point of connection with the utility. Programming Logic Controllers (PLCs) and analyzers to have dual power supply and to be backed up by emergency-backed UPS. Refer to the Global Facilities – Electrical – Electrical Specification Design Standard

5.10 CQC requirement

All parameters indicated in the Global Facilities – General – Design Parameters Best Known Methods (BKM) shall be monitored. Selection of analyzer shall meet the lower detection limit (LDL), sensitivity, and range required as per specification.
The parameters shall be monitored all the time, recorded in Micron SCADA system. Point of detection shall be at the outlet of the purifier or upstream of the distribution network where applicable as minimum.

5.11 Safety Requirement specific to H2 and O2

5.11.1 H_2

Civil and building requirement e.g. isolation wall, separation distance to be considered according to the local code and international guidelines. For hydrogen system setup, FM Global Property Loss Prevention Data Sheets 1-44 to be referenced to ensure relevant structure and requirement is met as per the datasheet on Damage Limiting Construction.

H_2 shall be distributed based on main pipeline(s) from the gas yard, building isolation cabinet(s) (BIC), and valve manifold box(es) (VMBs) equipped with their own controllers at subfab level.

The main pipeline(s) shall be equipped with a cut-off valve located at the gas yard or fab entrance. Depending on local codes and regulations, it shall be an AOV valve controlled by a Life Safety System (LSS) and provided with appropriate controls redundancy.

BICs and VMBs shall be configured with inlet pressure transducer process double isolation AOV/manual valves and purge port as a minimum. Additional stick features such as pressure regulators, pressure transducers, line filters, and vent out ports shall be evaluated during the design phase.

H2 shall be vented off to safe area. H_2 venting shall only be done when conditions exist where the H_2 diffusion in the surrounding air will not blow in the direction of the fab where H2 could be captured by the air handling units, thereby introducing H2 into the fab where it could alarm some hydride gas detection systems

5.11.2 O_2

The main pipeline(s) shall be equipped with a cut-off valve located at the gas yard or fab entrance. Depending on local codes and regulations, it shall be an AOV valve controlled by a Life Safety System (LSS) and provided with appropriate controls redundancy.

O_2 shall be vented off to safe area. O_2 venting shall only be done when conditions exist where the O_2 diffusion in the surrounding air will not blow in the direction of the fab where O_2 could be captured by the air handling units, thereby introducing high level of O_2 into the fab.

6.0 Document Control

6.1 **Approval**: GLOBAL_FAC_GFTT_DOC_APPR

6.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY

6.3 Final Viewer Access

All Micron sites Facilities Team Members

6.4 Retention

Adherence to the requirements specified in
Global Facilities - Document Control Procedures

6.5 Review

Biennially (two years)

7.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	10/29/20
1	Whole document Section 3 Section 5.5	Update the terms, GFTT and FCT as Global Facilties Engineering / Regional Team Discipline Removed acronyms the GFTT, FCT, GFC and descriptions for Global Facilities Planning, Central Team, Construction and Operations Team Updated Site Facilities Team Responsibilities Added a new section on bulk gas BCP guideline and criteria for greenfield projects	09/09/24

8.0 **Appendix 1**: Template Form for Site Specific Deviation List

Site: ______________________

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT